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STSR

Semi-independent policy training with shared representation for heterogeneous multi-agent RL

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The heterogeneity problem

One shared policy homogenizes

Parameter sharing works for identical agents, but pushes different unit types to behave alike.

Padding wastes compute

Zero-padding unifies input and output sizes, but adds redundant information and overhead.

Separate policies converge slowly

One network per agent avoids conflict, but costs more compute and converges slowly.

Goal: share knowledge where agents are alike, and specialize where they differ.

Three components of STSR

01

Common representation

A supervised encoder-decoder learns one input and output space that all agent types can share.

02

Semi-independent policies

Agents of the same type share one full policy. Different types share through hard parameter sharing.

03

Intrinsic rewards

Bonuses reward exploring unfamiliar states and keeping representations distinct across agent types.

Inspired by how the brain shares abstract knowledge, not only similar experiences.

Results on StarCraft II (SMAC)

Scenario	STSR rank
3s5z vs 3s6z	Best
MMM2	Best
3s5z	Second
27m vs 30m	Second

Ablation study

Every component contributed to performance.

Hard parameter sharing mainly speeds up training rather than raising final scores.

Baselines: VDN, QMIX, QPLEX, ROMA, RODE and CDS. Results are medians over 4 random seeds.

Transfer, limits and takeaways

Transfer and curriculum

Learned policies carried over to larger maps without retraining, especially when both sides grew.

Limitations

Less effective on homogeneous teams, where random clustering of identical agents sometimes did better.

Share where agents are alike, specialize where they differ.